# FUNDEMENTAL OF ALGORITHIM (CS502)

MUHAMMAD JUNAID

0304-1659294

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CS02 for final term important mcqs and Topic

Question No 1:

The greedy part of the Huffman encoding algorithm is to first find two nodes with **larger** frequency. (REP)

- True
- **False** **GOOGLE**

**Question No 2:**

Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles. (REP)

- True
- **False** **Pg159**

**Question No 3:**

Dijkestra s single source shortest path algorithm works if all edges weights are non-negative and there are no negative cost cycles. (REP)

- TRUE pg159
- FALSE

**Question No 4**

Dijkstra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles. (REP)

- True
- **False** **pg159**

**Question No 5:**

Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem. (REP)

- **True** **(p162)**
- False

**Question No 6:**

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for $d_{ij}^{(k)}$ (REP)

- **TRUE** **pg164**
- **FALSE**

**Question No 7:**

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The term coloring came from the original application which was in **map drawing** (REP)

- **True** **pg173**
- **False**

**Question No 8:**

The term "coloring" came from the original application which was in **architectural design** (REP)

- **True**
- **False** **pg173**

**Question No 9**

In the **clique cover** problem, for two vertices to be in the same group, they must be _____ each other. (REP)

- Apart from
- Far from
- Near to
- **Adjacent to** **(p176)**

**Question No 10**

In the clique cover problem, for two vertices to be in the same group, they **must be adjacent** to each other. (REP)

- True pg176
- False

Question No 11

In the **clique cover** problem, for two vertices to be in the same group, they **must be apart** from each other. (REP)

- True
- False pg176

Question No 12

The difference between Prims algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a different key. (REP)

- True pg156
- False

Question No 13

The difference between Prims algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key (REP)

- True pg156
- False

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Question No 14:

You have an adjective list for G, what is the time complexity to computer graph transpose G^T.? (REP)

V+E pg138

Question No 15

Given an adjacency list for G, it is possible to compute $G^T$ in $\Theta(V + E)$ time. (REP)

**It takes O(log V) to extract a vertex from the priority queue.**

Question No 16

Overall time for Kruskal is $\Theta(E \log E) = \Theta(E \log V)$ if the graph is sparse. (REP)

- True pg149
- False

Question No 17

An optimization problem is one in which you want to find, (REP)

- Not a solution
- An algorithm
- Good solution
- **The best solution pg197**

**Question No 18**

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices. (REP)

- **True Conceptual**
- **False**

Question No 19

If a graph has v vertices and e edges then to obtain a spanning tree we have to (REP)

- delete v edges.
- v – e + 5 edges
- v + e edges.
- **None of these**

Question No 20

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Huffman algorithm uses a greedy approach to generate an antefix code T that minimizes the expected length B (T) of the encoded string. (REP)

- True **False (p102)**

Question No 21

Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string. (REP)

True **False (p102)**

**Question No 22**

Huffman algorithm uses a greedy approach to generate a prefix code T that minimizes the expected length B (T) of of the encoded string. (REP)

True pg102 False

Question No 22

Bellman-Ford allows negative weights edges and negative cost cycles. True **False (p159)**

Question No 23

Bellman-Ford allows negative weight edges and no negative cost cycle True pg159
False

Question No 24

We do sorting to,

- keep elements in random positions
- keep the algorithm run in linear order
- keep the algorithm run in (log n) order
- **keep elements in increasing or decreasing order**

**Question No 25**

After partitioning array in Quick sort, pivot is placed in a position such that (REP)

- **Values smaller than pivot are on left and larger than pivot are on right**
- Values larger than pivot are on left and smaller than pivot are on right
- Pivot is the first element of array
- Pivot is the last element of array

**Question No 26**

Dynamic programming algorithms need to store the results of intermediate sub-problems. (REP)

- True pg75

- False

Question No 27

Which statement is true? (REP)

- If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal
- **If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.**
- **Both**
- **Nonne**

Question No 28

What **general property** of the list indicates that the graph has an **isolated vertex**? (REP)

- There is Null pointer at the end of list.
- **The Isolated vertex is not handled in list.** **GOOGLE**
- Only one value is entered in the list.
- There is at least one null list.

Question No 29

**Depth first search** is **shortest path** algorithm that works on **un-weighted** graphs. (REP)

- **True**
- **False**

**Question No 30**

Which is true statement? (REP)

- **Breadth first search is shortest path algorithm that works on un-weighted graphs (p153)**
- Depth first search is shortest path algorithm that works on un-weighted graphs.
- Both of above are true.
- None of above are true.

Question No 31

In Prim's algorithm, the additional information maintained by the algorithm is the **length of the shortest edge** from vertex v to points already in the tree. (REP)

- True
- **False**
- unknown

Question No 32

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Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a **large number of vertices**. (REP)

- **True**
- False
- unknown

Question No 33

If a problem is NP-complete, it must also be in NP. (REP)

**a) True** **pg 171**

b) False

c) unknown

Question No 34

Which statement is true? (REP)

(i) The running time of Bellman-Ford algorithm is T (VE)
(ii) Both Dijkstra's algorithm and Bellman-Ford are based on performing **repeated relaxations**
(iii) The 0-1 knapsack problem is hard to solve

- Only II
- Only III
- Both I and II
- All of the above

Question No 35

**What is generally true of Adjacency List and Adjacency Matrix representations of graphs?** (REP)

- Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)
- **Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2) (page 116)**
- Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)
- Lists require more space than matrices but are faster to find the weight of an edge (v1,v2)

Question No 36

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**If a graph has v vertices and e edges then to obtain a spanning tree we have to delete** (REP)

- v edges.
- v – e + 5 edges
- v + e edges.
- **None of these(page 125)**

Question No 37

**The Huffman algorithm finds a (n) _____ solution.** (REP)

- ► **Optimal** **pg105**
- ► Non-optimal
- ► Exponential
- ► Polynomial

Question No 38

**The Huffman algorithm finds an exponential solution** (REP)

- True
- **False** **pg105**

Question No 39

**The Huffman algorithm finds a polynomial solution** (REP)

- True
- False

Question No 40

If there were a polynomial solution for even a single NP-complete problem, then ever problem in NPC will be solvable in polynomial time. (REP)

- True pg173
- False

Question No 41

The class NP-complete (NPC) problems consists of a set of decision problems (a subset of class NP) that no one knows how to solve (REP)

- Polynomial
- Optimal
- Efficiently pg173
- None

Question No 42

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**The greedy part of the Huffman encoding algorithm is to first find two nodes with larger frequency.** (REP)

- True
- **False** **(page 100)**

Question No 43

**The code word assigned to characters by the Huffman algorithm have the property that no code word is the postfix of any other.** (REP)

- **True**
- **False**

**Question No 44:**

**Shortest path problems can be solved efficiently by modeling the road map as a graph.**

- **False**
- **True**

**Question No 45**

**Dijkestra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.** (REP)

- True
- **False** **( page 154)**

**Question No 46**

**Dijkstra's algorithm is operates by maintaining a subset of vertices** (REP)

- TRUE Not Sure
- False

**Question No 47**

**What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?** (REP)

- **O log (v)** **( page 152)**
- v.v
- e.e
- log(e)

**Question No 48**

**You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?** (REP)

- **V+e**
- **V.e** **pg138**
- **V**
- **e**

Question No 49

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►Which of the following arrays represent descending (max) heaps? (REP)

I.

[10,7,7,2,4,6]

II.

[10,7,6,2,4,7]

III.

[10,6,7,2,4,6]

IV.

[6,6,7,2,4,10]

- **Only II (page-140)**
- Only IV
- Both II and IV
- Both I and III

Question No 50

Cross edge is

- (u,v) where u and v are not ancestor of one another
- (u,v) were u is ancestor of v and v is not descendent of u
- (u,v) were u is not ancestor of v and v is t descendent of u pg129
- (u,v) were u and v either is ancestor or descendent

Question No 51

According to parenthesis lemma vertex u is a ancstor of v vertex if and only if:

f [d[u], f[u]] ⊇ [d[v], f[v]] Page129 .

The ancestor and descendent relation can be nicely inferred by the *parenthesis* lemma. u is a descendent of v if and only if [d[u], f[u]] ⊆ [d[v], f[v]]. u is a ancestor of v if and only if [d[u], f[u]] ⊇ [d[v], f[v]]. u

**Question No 52**

Accordig to parenthesis lemma vertex u is a unrelated of v vertex if and only if

[d[u], f[u]] and [d[v], f[v]] Page 129

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The ancestor and descendent relation can be nicely inferred by the *parenthesis* lemma. u is a descendent of v if and only if $[d[u], f[u]] \subseteq [d[v], f[v]]$. u is a ancestor of v if and only if $[d[u], f[u]] \supseteq [d[v], f[v]]$. u is unrelated to v if and only if $[d[u], f[u]]$ and $[d[v], f[v]]$ are disjoint. The is shown in Figure 8.26. The width of the rectangle associated with a vertex is equal to the time the vertex was discovered till the time

**Question No 53**

For undirected graphs, there is no distinction between_____ and _____ edges

**Forward and Backward** **Page 130**

For *undirected* graphs, there is no distinction between forward and back edges. By convention they are all called back edges. Furthermore, there are no cross edges (can you see why not?)

**Question No 54:**

Bellman-Ford applies relaxation to every _____ of the graph and repeats this V − 1 times

Edge

Like Dijkstra's algorithm, Bellman-Ford is based on performing repeated relaxations. Bellman-Ford applies relaxation to *every edge* of the graph and repeats this V − 1 times. Here is the algorithm; its is

**Question No 55**

The tricky part of _____ algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A.

**Kruskal's (Page 147)**

Kruskal's algorithm works by adding edges in increasing order of weight (lightest edge first). If the next edge does not induce a cycle among the current set of edges, then it is added to A. If it does, we skip it and consider the next in order. As the algorithm runs, the edges in A induce a *forest* on the vertices. The trees of this forest are eventually merged until a single tree forms containing all vertices.

The tricky part of the algorithm is how to detect whether the addition of an edge will create a cycle in A. Suppose the edge being considered has vertices $(u, v)$. We want a fast test that tells us whether u and v are in the same tree of A. This can be done using the *Union-Find* data structure which supports the following $O(\log n)$ operations:

**Question No 56**

Finding the faster result of the shortest path from u to v for every pair of vertices u and v we use_____.

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**All-pairs shortest-paths problem (Page 154)**

**All-pairs shortest-paths problem:** Find a shortest path from $u$ to $v$ for *every pair* of vertices $u$ and $v$. Although this problem can be solved by running a single-source algorithm once from each vertex, it can usually be solved faster.

**Question No 57**

Edge weights can be interpreted as distance _____.

**in the shortest-paths (Page 153)**

**Question No 58**

Dijkstra's Algorithm is used to solve_____problems.

**Single-source shortest path (Page 154)**

### 8.6.1 Dijkstra's Algorithm

Dijkstra's algorithm is a simple *greedy* algorithm for computing the **single-source shortest-paths** to all other vertices. Dijkstra's algorithm works on a weighted directed graph $G = (V, E)$ in which all edge weights are non-negative, i.e., $w(u, v) \geq 0$ for each edge $(u, v) \in E$.

**Question No 59**

Problems such as the shortest route between cities can be solved efficiently by modeling the road map as a_____.

**Graph (Page 153)**

Problems such as shortest route between cities can be solved efficiently by modelling the road map as a graph. The nodes or vertices represent cities and edges represent roads. Edge weights can be interpreted as distances. Other metrics can also be used, e.g., time, cost, penalties and loss.

**Question No 60**

For _____ graphs, there is no distinction between forward and back edges.

**Undirected (Page 130)**

**Question No 61**

Bellman-Ford Algorithm does not allow G(graph) to have _____.

**negative cost cycles (Page 159)**

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Dijkstra's single-source shortest path algorithm works if all edges weights are non-negative and there are no negative cost cycles. Bellman-Ford *allows* negative weights edges and no negative cost cycles. The algorithm is slower than Dijkstra's, running in $\Theta(VE)$ time.

**Question No 62**

A topological sort of a DAG is a _____ ordering of the vertices of the DAG such that for each edge (u, v), u appears before v in the ordering.

**Linear (Page 133)**

A topological sort of a DAG is a linear ordering of the vertices of the DAG such that for each edge $(u, v)$, u appears before v in the ordering.

**Question No 63**

Networks are _____ in the sense that it is possible from any location in the network to reach any other location in the digraph.

**Complete (Page 135)**

We consider an important connectivity problem with digraphs. When diagraphs are used in communication and transportation networks, people want to know that their networks are *complete*. Complete in the sense that that it is possible from any location in the network to reach any other location in the digraph.

**Question No 64**

According to parenthesis lemma vertex u is a descendent of v vertex if and only if:\

**f [d[u], f[u]] ⊆ [d[v], f[v]] (Page 129)**

The ancestor and descendent relation can be nicely inferred by the *parenthesis* lemma. u is a descendent of v if and only if $[d[u], f[u]] \subseteq [d[v], f[v]]$. u is a ancestor of v if and only if $[d[u], f[u]] \supseteq [d[v], f[v]]$. u

**Question No 65**

For _____ graphs, there is no distinction between forward and back edges.

**Undirected (Page 130)**

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For *undirected* graphs, there is no distinction between forward and back edges. By convention they are all called back edges. Furthermore, there are no cross edges (can you see why not?)

For *undirected* graphs, there is no distinction between forward and back edges. By convention they are all called back edges. Furthermore, there are no cross edges (can you see why not?)

**Question No 66**

There exist a unique path between any _____ vertices of a free tree.

**Two (Page 142)**

unique path between any two vertices of a free tree. Adding any edge to a free tree creates a unique cycle. Breaking any edge on this cycle restores the free tree. This is illustrated in Figure 8.45. When the edges (b, e) or (b, d) are added to the free tree, the result is a cycle.

**Question No 67**

Keeping in mind the shortest-path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are_____and Edges are_____which may be wired or wireless.

**Routers, communication links (Page 153)**

Similar scenarios occur in computer networks like the Internet where data packets have to be routed. The vertices are *routers*. Edges are communication links which may be be wire or wireless. Edge weights can be distance, link speed, link capacity link delays, and link utilization.

**Question No 68**

In undirected graph, by convention all the edges are called _____ edges.

**back (Page 130)**

For *undirected* graphs, there is no distinction between forward and back edges. By convention they are all called back edges. Furthermore, there are no cross edges (can you see why not?)

**Question No 69**

Bellman-Ford algorithm is slower than_____.

**Dijkstra's (Page 159)**

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Dijkstra's single-source shortest path algorithm works if all edges weights are non-negative and there are no negative cost cycles. Bellman-Ford *allows* negative weights edges and no negative cost cycles. The algorithm is slower than Dijkstra's, running in $\Theta(VE)$ time.

**Question No 70**

In the shortest-paths problem, we are given a weighted of_____G = (V, E).

**directed graph Page 129**

In the *shortest-paths problem* We are given a weighted, directed graph $G = (V, E)$ The weight of path $p = < v_0, v_1, \ldots, v_k >$ is the sum of the constituent edges:

**Question No 71**

If, in a DFS forest of digraph G = (V, E), f[u] = f[v] for an edge (u, v) ? E then the edge is called

- **Back edge (page 129)**
- Forward edge
- Cross Edge
- Tree Edge

Question No 72

In inductive approach of knapsack problem, we consider 2 cases, _____Or _____.

Leave object, Take object

Question No 73

In an undirected graph, we say that an edge is _____on a vertex if the vertex is an endpoint of the edge.

Incident

Question No 74

There are _____ ways of representing graphs.

2

**Question No 75**

The number of edges that come out of a vertex is called the _____ of that vertex in a digraph.

out-degree

**Question No 76**

In general, the Activity Selection Problem is to select a _____

Maximum-size set of mutually non-interfering activities

**Question No 78**

Brute-force is a simple strategy for computing _____.

shortest path

**Question No 79**

Which of the following problems is NOT solved using dynamic programming?

Fractional knapsack problem

**Question No 80**

If a matrix has three rows and two columns, then dimensions of matrix will be:

3 x 2

**Question No 81**

If the graph is represented using an adjacency matrix, then Breadth-first search takes _____ time.

O(V^2)

**Question No 83**

The general coin change problem can be solved using_____.

Dynamic programming

**Question No 84**

If there are no cycles in the graph, what is the maximum number of edges present in a simple directed graph with 5 vertices?

5

**Question No 85**

Traversing a graph means visiting _____ in the graph.

every node

**Question No 86**

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain _____.

vi (Value of object i)

**Question No 87**

In general, the activity selection problem is to select a _____.

maximum-size set of mutually non-interfering

activities

**Question No 88**

_____ data structure is used to perform the Depth-first search.

Stack

**Question No 89**

In the context of encoding/decoding, fixed-length codes may not be efficient from the perspective of _____the total quantity of data.

Minimizing

**Question No 90**

In general, a graph G = (V,E) consists of a _____ and E, a binary relation on V called edges.

Finite set of vertices V

**Question No 90**

If the string "lmncde" is coded with ASCII code, the message length would be _____ bits.

60

**Question No 91**

Fractional Knapsack is founded on _____ method.

Greedy

**Question No 92**

In an undirected graph, we say that an edge is _____on a vertex if the vertex is an endpoint of the edge.

Incident

**Question No 93**

The running time of the traversal algorithm depends on _____and _____ which is used for the bag.

Graph representation, Data structure

**Question No 94**

The greedy approach gives us an optimal solution when the coins are all powers of a _____denomination.

Fixed

**Question No 95**

In general, a graph G = (V,E) consists of a _____ and E, a binary relation on V called edges.

Finite set of vertices V

Question No 96

Brute-force is a simple strategy for computing _____.

shortest path

**Question No 97**

If there are no cycles in the graph, what is the maximum number of edges present in a simple directed graph with 5 vertices?

5

**Question No 98**

Bag is a _____

data structure

**Question No 99**

In Dynamic Programming based solution of knapsack problem, to compute entries of 'V', we will imply a(n) _____ approach.

Inductive

**Question No 100**

Using ASCII standard the string "greedy" will be encoded with

48 bits

**Question No 101**

In Activity scheduling algorithm, the width of a rectangle _____

Indicates the duration of an activity

**Question No 102**

The general coin change problem can be solved using_____.

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Dynamic programming

**Question No 103**

How many steps are involved to design the dynamic programming strategy?

4

**Question No 104**

In Dynamic Programming based solution of knapsack problem, to compute entries of 'V', we will imply a(n) _____ approach.

Inductive

**Question No 105**

In Fractional Knapsack problem, one is allowed to take fraction of an item for _____

Both, fraction of the weight and value

**Question No 106**

In Huffman encoding, the _____ is the number of occurrences of a character divided by the total characters in the message.

Parsing

**Question No 107**

In Knapsack Problem, each item must be entirely accepted or rejected, is called _____ problem.

0-1

**Question No 108**

_____ is a good way to model some sort of "connection" or "relationship" or "interaction" between pairs of objects taken from a set of objects.

Graph

**Question No 109**

In Huffman encoding fixed-length codes are popular because it is very easy to break up a_____ into its individual_____.

String, Characters

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**Question No 110**

There are _____ ways of representing graphs.

2

**Question No 111**

The probability in Huffman encoding is the number of occurrences of a character divided by the total _____in the message.

Characters

**Question No 112**

Queue data structure work on _____ principles.

FIFO (first in, first out)

Question No 113

A free tree with n vertices has exactly _____edges.

- n
- n+1

**n-1 page 142**

- 1

Question No 114

Kruskal`s algorithm works by adding _____ in increasing order of weight (lightest edge first)

- Verticals
- **Edges page 147**
- Trees
- Weights

Question No 115

Computing the strongly connected components of a digraphs is a/an _____ of the problem to determine whether a diagraph is strongly connected or not

- Size
- **Generalization page 135**
- Optimization
- Connection

Question No 116:

The process of updating estimates in Dijkstra`s algorithm is called _____

- Updating

- Amendment
- **Relaxation page 154**
- Insertion

Question No 117

An un-weighted graph can be considered as a graph in which every edge has weight_____ Unit.

- 7
- 5
- 3
- **1 page 153**

**Question No 118**

The breadth-first-search algorithm is a shortest-path algorithm that works on_____ graphs.

- Weighted
- Directed
- **Un-weighted page 153**
- Un-directed

Question No 119

Which activity creates a unique cycle in a free tree:

- adding any vertex
- adding any sub tree
- adding root
- **adding any edge page 142**

Question No 120

An edge (u, v) _____ E - A is safe if A _____ {(u, v)} is viable.

- **Ɛ,U page 143**
- U,Ɛ
- Ɛ,Π
- Π,U

Question No 121

The relationship between number of back edges and Number of cycles in DFS is,

a. Both are equal

b. Back edges are half of cycles

c. Back edges are one quarter of cycles

**d. There is no relationship between no. of edges and cycles go ogle**

Question No122

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12. Dijkstra's Algorithm is used to solve_____ problems.

- ► Multi-source shortest path
- ► Sorting & searching
- ► All-Pair shortest path
- ► Single-Source Shortest Path

Question No 123

A digraph is strongly connected under what condition?

- ► A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v.
- ► **A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa. Page 135**
- ► A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.
- ► A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

Question No 124

Floyd-Warshall Algorithm is based on _____

- o **Dynamic Programming page 161**
- o Greedy Approach
- o Divide and Conquer
- o Complexity theory

Question No 125

In strong components algorithm, vertices are sorted in _____order of finish times.

- o Any
- o Increasing
- o Decreasing pg141
- o Strong

Question No126

_____ is commonly the running time of Dijkstra's Algorithm using the binary heap method.

- o **Θ (E log V) page 156**
- o Θ (V log )
- o Θ (log E)
- o Θ (Blog V)

Question No 127

Which technique is used in the implementation of Kruskal solution for the MST?

- o **Greedy Technique page 142 / google**
- o Divide-and-Conquer Technique

- Dynamic Programming Technique
- The algorithm combines more than one of the above techniques i.e. Divide-and-Conquer and Dynamic Programming

Question No 128

A fully connected undirected graph of 5 nodes will have _____ edges.

- 4
- 5
- 10 n(n-1)/2
- 15

Question N0 129

Prim's algorithm is based on_____ strategy.

- **Greedy page 150**
- Dynamic programming
- Divide and Conquer
- Exponential

Question No 130

We say that two vertices u and v are mutually _____if u can reach v and vice versa.

- Crossed
- Forward
- **Reachable page 135**
- Not Reachable

Question No 131

A strongly connected component only apply to:

- Directed Graph page 135
- Undirected Graph
- Minimum Spanning Tree
- Breadth First Search

Question No 132

A graph may contain_____

- Exactly one MST
- No MST
- One or zero MST

**More than one MST google**

Question No 132

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In _____ algorithm(s), at any time, the subset of edges A forms a single tree.

- Kurskal's
- Prims's pg149
- kruskal's and Prim's
- BFS

## TOPIC


- Knapsack Problem pg90
- dynamic programming strategy pg92
- Huffman Encoding pg100
- Prefix Property:
- Graph pg113
- Breadth First Search pg117
- Depth First search pg119
- Generic Graph Traversal Algorithm pg120
- Timestamp Structure pg125
- Edge Graph Classification pg128
- Topological Sort pg133
- Kruskal's Algorithm pg147
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- Few variants of the shortest path problem pg153
- Dijkstra's Algorithm pg154
- Bellman-Ford Algorithm pg159
- Complexity Theory pg169
- Coplexicity class pg170